

GPU Acceleration in Optimization and Optimal Control

Pre-Congress Workshop, 23rd IFAC World Congress

Sungho Shin

shin.mit.edu

Massachusetts Institute of Technology

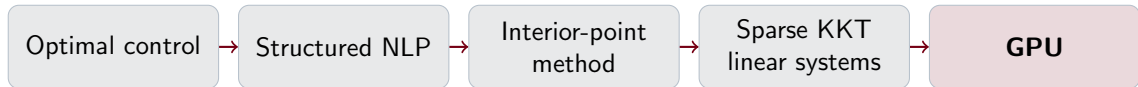
23 August 2026 BEXCO, Busan

Logistics

Time	Part	Topic
08:35–10:20	Lecture	Part I: nonlinear optimization Part II: optimal control as a structured NLP Part III: GPU acceleration
10:20–10:40	<i>coffee break</i>	
10:40–11:15	Hands-on 1	GPU computing in Julia
11:15–11:55	Hands-on 2	Optimal control and parameter estimation on GPUs

► Slides and notebooks: <https://madsuite.org/ifac2026/>

Scope and Roadmap



- ▶ **Part I:** nonlinear optimization; optimality conditions and the interior-point method.
- ▶ **Part II:** optimal control as a structured nonlinear program; transcription and collocation.
- ▶ **Part III:** GPU architecture, sparse linear algebra on GPUs, condensed-space interior-point methods.
- ▶ In the hands-on session after the break you run the whole pipeline yourself.

Table of Contents

Part I

Nonlinear optimization

1. Nonlinear Programs
2. Optimality Conditions
3. Interior-Point Methods
4. The Software Stack
5. Summary

Part II

Optimal control

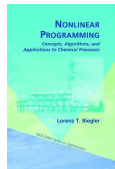
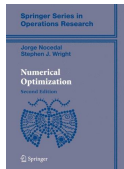
6. From Optimal Control to Nonlinear Programs
7. Collocation
8. Summary

Part III

GPU acceleration

9. A GPU Primer for Optimizers
10. Why NLP on GPUs Is Hard
11. SIMD Abstraction: ExaModels.jl
12. Pivoting-Free Linear Algebra
13. Condensed-Space Interior-Point Methods
14. Numerical Results
15. Variants and Outlook

References



- ▶ **Nocedal and Wright (2006)**: the standard reference; Newton (Ch. 3), AD (Ch. 8), optimality (Ch. 12), interior point (Ch. 19).
- ▶ **Biegler (2010)**: optimal control and process systems; Ch. 10 is the source for Part II.
- ▶ **Wächter and Biegler (2006)**: the Ipopt paper; Part I's algorithm in detail.
- ▶ **Hands-on software**: ExaModels.jl, MadNLP.jl (Shin, 2025; Shin, Pacaud, et al., 2025).

Part I

Nonlinear Optimization

Optimality conditions, interior-point methods, the software stack

Outline

- ▶ **Part I: Nonlinear Optimization**
 - ▶ Nonlinear Programs
 - ▶ Optimality Conditions
 - ▶ Interior-Point Methods
 - ▶ The Software Stack
 - ▶ Summary
- ▶ Part II: Optimal Control as a Structured NLP
- ▶ Part III: GPU Acceleration
- ▶ Part IV: Hands-On Sessions

Nonlinear Program (NLP) = Nonlinear Optimization

$$\min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x})$$

$$\text{s. t. } \mathbf{h}(\mathbf{x}) = \mathbf{0} \quad (\text{equality constraints, } m_E \text{ of them})$$

$$\mathbf{g}(\mathbf{x}) \leq \mathbf{0} \quad (\text{inequality constraints, } m_I \text{ of them})$$

with $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $\mathbf{h} : \mathbb{R}^n \rightarrow \mathbb{R}^{m_E}$, $\mathbf{g} : \mathbb{R}^n \rightarrow \mathbb{R}^{m_I}$ twice continuously differentiable.

- ▶ **Nonconvex**, and we look only for a **local solution**: a feasible point no nearby feasible point improves on. Global optimality is not our concern here.
- ▶ **Large, but structured**. Discretizing an optimal control problem makes n big; what comes out is **sparse and repetitive**, and exploiting that structure is what makes it solvable.

Outline

► Part I: Nonlinear Optimization

- Nonlinear Programs
- Optimality Conditions
- Interior-Point Methods
- The Software Stack
- Summary

► Part II: Optimal Control as a Structured NLP

► Part III: GPU Acceleration

► Part IV: Hands-On Sessions

From Local Solutions to Optimality Conditions

What we are looking for

$\mathbf{x}^*$ is a **local solution** if it is feasible and no feasible point near it is better:

$$\exists \varepsilon > 0 : \quad f(\mathbf{x}^*) \leq f(\mathbf{x}) \quad \text{for every feasible } \mathbf{x} \text{ with } \|\mathbf{x} - \mathbf{x}^*\| < \varepsilon.$$

- ▶ **An algorithm cannot check that.** It quantifies over every nearby feasible point, while a solver holds one point and its derivatives.
- ▶ **Optimality conditions** replace it with equations and inequalities in f , $\mathbf{h}$, $\mathbf{g}$ and their derivatives **at a single point**, which is checkable and something a solver can drive to zero.

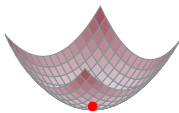
Plan

unconstrained $\rightarrow$ equality constrained $\rightarrow$ inequality constrained $\rightarrow$ KKT

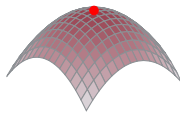
Unconstrained Problems

At a local solution $\mathbf{x}^*$ of $\min_{\mathbf{x}} f(\mathbf{x})$, from the Taylor expansion:

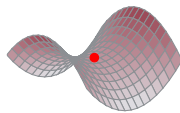
- ▶ First-order necessary: $\nabla f(\mathbf{x}^*) = \mathbf{0}$.
- ▶ Second-order necessary: $\nabla^2 f(\mathbf{x}^*) \succeq \mathbf{0}$.
- ▶ Second-order sufficient: $\nabla f(\mathbf{x}^*) = \mathbf{0}$ and $\nabla^2 f(\mathbf{x}^*) \succ \mathbf{0}$ imply a local solution.



local min



local max

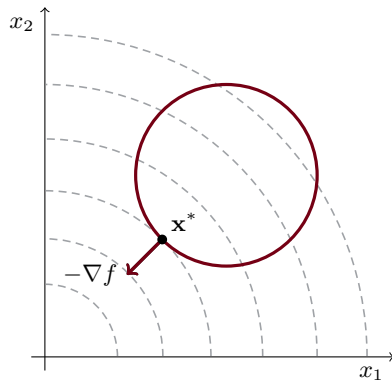


saddle point

- ▶ The second-order condition is **computationally difficult to check**, so most algorithms aim for a **stationary point**, $\nabla f(\mathbf{x}^*) = \mathbf{0}$, and rely on **globalization** strategies that reduce the objective at each step.

Equality, Active, and Inactive Constraints

$$\min_{x_1, x_2} f(x_1, x_2) \quad \text{s. t. } h(x_1, x_2) = 0$$

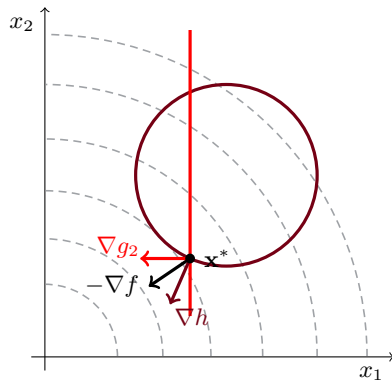


- **Equality:** feasible moves stay on the curve, so at a solution $-\nabla f$ lies along ∇h .

$$-\nabla f = \lambda \nabla h, \quad \lambda \text{ free}$$

Equality, Active, and Inactive Constraints

$$\min_{x_1, x_2} f(x_1, x_2) \quad \text{s. t. } h(x_1, x_2) = 0, \quad g_2(x_1, x_2) \leq 0$$

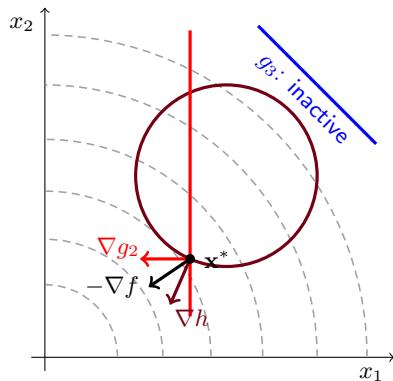


- **Equality**: feasible moves stay on the curve, so at a solution $-\nabla f$ lies along ∇h .
- **Active** g_2 , meaning $g_2(\mathbf{x}^*) = 0$: it binds, the solution moves, and ∇g_2 joins, with **one sign** only, since g_2 can push back but not pull.

$$-\nabla f = \lambda \nabla h + \mu_2 \nabla g_2, \quad \mu_2 \geq 0$$

Equality, Active, and Inactive Constraints

$$\min_{x_1, x_2} f(x_1, x_2) \quad \text{s. t. } h(x_1, x_2) = 0, \quad g_2(x_1, x_2) \leq 0, \quad g_3(x_1, x_2) \leq 0$$



- **Equality**: feasible moves stay on the curve, so at a solution $-\nabla f$ lies along ∇h .
- **Active** g_2 , meaning $g_2(\mathbf{x}^*) = 0$: it binds, the solution moves, and ∇g_2 joins, with **one sign** only, since g_2 can push back but not pull.
- **Inactive** g_3 ($g_3(\mathbf{x}^*) < 0$): no effect, and which constraints are active is **not known in advance**.

$$-\nabla f = \lambda \nabla h + \mu_2 \nabla g_2 + \mu_3 \nabla g_3$$

$$\mu_2, \mu_3 \geq 0, \quad \mu_i g_i(\mathbf{x}^*) = 0 \Rightarrow \mu_3 = 0$$

The Lagrangian and the Karush–Kuhn–Tucker Conditions

$$\min_{\mathbf{x}} f(\mathbf{x}) \quad \text{s. t.} \quad \mathbf{h}(\mathbf{x}) = \mathbf{0}, \quad \mathbf{g}(\mathbf{x}) \leq \mathbf{0}$$

Both enter the *Lagrangian* $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = f + \boldsymbol{\lambda}^\top \mathbf{h} + \boldsymbol{\mu}^\top \mathbf{g}$; only the sign of $\boldsymbol{\mu}$ differs.

KKT conditions: at a local solution $\mathbf{x}^*$, *under a constraint qualification*, there exist $\boldsymbol{\lambda}^*, \boldsymbol{\mu}^*$ with

$$\nabla f(\mathbf{x}^*) + \nabla \mathbf{h}(\mathbf{x}^*) \boldsymbol{\lambda}^* + \nabla \mathbf{g}(\mathbf{x}^*) \boldsymbol{\mu}^* = \mathbf{0} \quad (\text{stationarity})$$

$$\mathbf{h}(\mathbf{x}^*) = \mathbf{0}, \quad \mathbf{g}(\mathbf{x}^*) \leq \mathbf{0} \quad (\text{primal feasibility})$$

$$\boldsymbol{\mu}^* \geq \mathbf{0} \quad (\text{dual feasibility})$$

$$\mu_i^* g_i(\mathbf{x}^*) = 0 \quad (\text{complementary slackness})$$

- ▶ Second order: $\mathbf{w}^\top \nabla_{\mathbf{xx}}^2 \mathcal{L} \mathbf{w} > 0$ on the tangent space of the active constraints.
- ▶ These conditions are **checkable at a single point**, which is what makes them usable by an algorithm. The next section drives them to zero.

The Combinatorial Complexity of Inequality Constraints

$$\min_{\mathbf{x}} f(\mathbf{x}) \quad \text{s. t.} \quad \mathbf{h}(\mathbf{x}) = \mathbf{0}, \mathbf{g}(\mathbf{x}) \leq \mathbf{0}$$

- ▶ Which inequalities are active is not known until the problem is solved: at $\mathbf{x}^*$ each g_i is tight or slack, and 2^{m_I} active sets are possible.
- ▶ Active-set methods guess the active set and solve an equality-constrained subproblem. Excellent when small, and they warm-start well; the number of updates grows badly with size.
- ▶ Interior-point methods *replace* complementarity by a smooth approximation, and solve a sequence of smooth problems. The method of choice at these sizes, and the one that maps onto a GPU.

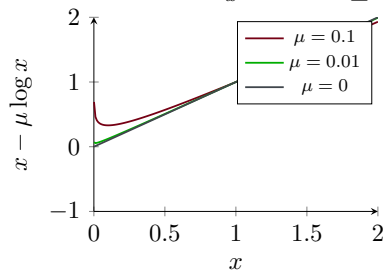
The Barrier Subproblem

$$\min_{\mathbf{x}} f(\mathbf{x}) \quad \text{s. t. } \mathbf{h}(\mathbf{x}) = \mathbf{0}, \mathbf{x} \geq \mathbf{0}$$

⇓ replace the bound by a barrier

$$\min_{\mathbf{x}} f(\mathbf{x}) - \mu \sum_i \log x_i \quad \text{s. t. } \mathbf{h}(\mathbf{x}) = \mathbf{0}$$

the scalar case $\min_x x \text{ s. t. } x \geq 0$



- ▶ The barrier $-\mu \log x_i$ blows up at the boundary, so any iterate with a finite objective has $\mathbf{x} > \mathbf{0}$ automatically.
- ▶ **The inequality is gone:** this is an equality-constrained problem, whose optimality conditions we already have.
- ▶ The minimizer of $x - \mu \log x$ over $x > 0$ is $x = \mu$, so as $\mu \downarrow 0$ it slides to the boundary and the barrier solution approaches a solution of the original problem.

The Homotopy Interpretation

Writing the first-order conditions of the barrier subproblem and introducing $\mathbf{z} = \mu \mathbf{X}^{-1} \mathbf{1}$ as the bound multipliers gives the **perturbed KKT system**:

$$\nabla f(\mathbf{x}) + \nabla \mathbf{h}(\mathbf{x}) \boldsymbol{\lambda} - \mathbf{z} = \mathbf{0}$$

$$\mathbf{h}(\mathbf{x}) = \mathbf{0}$$

$$\mathbf{XZ}\mathbf{1} = \mu \mathbf{1} \quad (\leftarrow \text{this is the smoothing})$$

where $\mathbf{X} = \text{diag}(\mathbf{x})$, $\mathbf{Z} = \text{diag}(\mathbf{z})$.

- ▶ At $\mu = 0$ these are the KKT conditions. For $\mu > 0$ it is a *smooth, square* nonlinear system with a unique solution $(\mathbf{x}(\mu), \boldsymbol{\lambda}(\mu), \mathbf{z}(\mu))$ under standard assumptions.
- ▶ The set $\{(\mathbf{x}(\mu), \boldsymbol{\lambda}(\mu), \mathbf{z}(\mu)) : \mu > 0\}$ is the **central path**.
- ▶ **The algorithm**: follow the central path with Newton steps, driving μ down to a very small value ($\mu \approx 10^{-9}$) and **never to 0**.

The Newton Step

Newton's method on the perturbed KKT system of the previous slide, at the iterate $(\mathbf{x}_k, \boldsymbol{\lambda}_k, \mathbf{z}_k)$:

$$\begin{bmatrix} \nabla_{\mathbf{x}\mathbf{x}}^2 \mathcal{L}(\mathbf{x}_k, \boldsymbol{\lambda}_k) & \nabla \mathbf{h}(\mathbf{x}_k) & -\mathbf{I} \\ \nabla \mathbf{h}(\mathbf{x}_k)^\top & \mathbf{0} & \mathbf{0} \\ \mathbf{Z}_k & \mathbf{0} & \mathbf{X}_k \end{bmatrix} \begin{bmatrix} \mathbf{d}_k^{\mathbf{x}} \\ \mathbf{d}_k^{\boldsymbol{\lambda}} \\ \mathbf{d}_k^{\mathbf{z}} \end{bmatrix} = - \begin{bmatrix} \nabla f(\mathbf{x}_k) + \nabla \mathbf{h}(\mathbf{x}_k) \boldsymbol{\lambda}_k - \mathbf{z}_k \\ \mathbf{h}(\mathbf{x}_k) \\ \mathbf{X}_k \mathbf{Z}_k \mathbf{1} - \mu_k \mathbf{1} \end{bmatrix}$$

- ▶ Three equations, three unknowns: one Newton system per iteration.
- ▶ The last row is **diagonal**, so $\mathbf{d}_k^{\mathbf{z}}$ comes out exactly:

$$\mathbf{d}_k^{\mathbf{z}} = -\mathbf{z}_k + \mu_k \mathbf{X}_k^{-1} \mathbf{1} - \mathbf{X}_k^{-1} \mathbf{Z}_k \mathbf{d}_k^{\mathbf{x}}.$$

- ▶ Substituting it into the first row leaves a smaller system in $(\mathbf{d}_k^{\mathbf{x}}, \mathbf{d}_k^{\boldsymbol{\lambda}})$ alone, treated in the next slide.

The KKT Matrix

With $\mathbf{d}^z$ eliminated, this is what is solved at **every** iteration:

$$\underbrace{\begin{bmatrix} \nabla_{\mathbf{x}\mathbf{x}}^2 \mathcal{L}(\mathbf{x}_k, \boldsymbol{\lambda}_k) + \mathbf{X}_k^{-1} \mathbf{Z}_k & \nabla \mathbf{h}(\mathbf{x}_k) \\ \nabla \mathbf{h}(\mathbf{x}_k)^\top & \mathbf{0} \end{bmatrix}}_{\mathbf{K}} \begin{bmatrix} \mathbf{d}_k^{\mathbf{x}} \\ \mathbf{d}_k^{\boldsymbol{\lambda}} \end{bmatrix} = \begin{bmatrix} -\nabla f(\mathbf{x}_k) - \nabla \mathbf{h}(\mathbf{x}_k) \boldsymbol{\lambda}_k + \mu_k \mathbf{X}_k^{-1} \mathbf{1} \\ -\mathbf{h}(\mathbf{x}_k) \end{bmatrix}$$

- **Blue**: the barrier contribution. As $\mu \rightarrow 0$ the entries of $\mathbf{X}^{-1} \mathbf{Z}$ split into $O(\mu)$ and $O(1/\mu)$: the system becomes **increasingly ill-conditioned**, by construction.
- The matrix is large, **sparse**, and **symmetric indefinite**. Its sparsity pattern is fixed across iterations; only the values change.

The KKT Matrix

With $\mathbf{d}^z$ eliminated, this is what is solved at **every** iteration:

$$\underbrace{\begin{bmatrix} \nabla_{\mathbf{x}\mathbf{x}}^2 \mathcal{L}(\mathbf{x}_k, \boldsymbol{\lambda}_k) + \mathbf{X}_k^{-1} \mathbf{Z}_k + \delta_k \mathbf{I} & \nabla \mathbf{h}(\mathbf{x}_k) \\ \nabla \mathbf{h}(\mathbf{x}_k)^\top & -\gamma_k \mathbf{I} \end{bmatrix}}_{\mathbf{K}(\delta_k, \gamma_k)} \begin{bmatrix} \mathbf{d}_k^{\mathbf{x}} \\ \mathbf{d}_k^{\boldsymbol{\lambda}} \end{bmatrix} = \begin{bmatrix} -\nabla f(\mathbf{x}_k) - \nabla \mathbf{h}(\mathbf{x}_k) \boldsymbol{\lambda}_k + \mu_k \mathbf{X}_k^{-1} \mathbf{1} \\ -\mathbf{h}(\mathbf{x}_k) \end{bmatrix}$$

- ▶ **Red**: the two regularization knobs. The Newton direction is productive only when $\text{Inertia}(\mathbf{K}) = (n, m_E, 0)$.
- ▶ **Inertia correction**: from $\delta_k = \gamma_k = 0$, factorize $\mathbf{K}$; if the inertia is right, take the step; otherwise raise the **primal** δ_k , set the **dual** $\gamma_k > 0$, and factorize again.
- ▶ **One factorization per trial**, every iteration. The factorization reports the inertia **for free**, so the loop costs no more than the factorizations themselves.

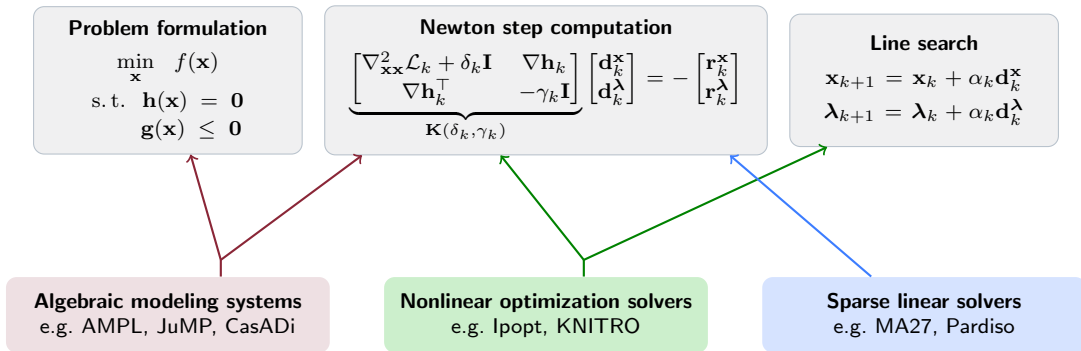
The Interior-Point Algorithm

Primal–dual interior-point method

Given $\mathbf{x}_0 > \mathbf{0}$, $\mathbf{z}_0 > \mathbf{0}$, $\mu_0 > 0$:

1. Evaluate $f, \mathbf{h}$ and their derivatives; assemble the KKT matrix.
2. **Factorize** $\mathbf{K}(\delta_k, \gamma_k)$, correcting the inertia with δ_k , and **solve** with those factors for the step: $\mathbf{K}(\delta_k, \gamma_k) [\mathbf{d}_k^{\mathbf{x}}; \mathbf{d}_k^{\lambda}] = -\mathbf{r}_k$. The linear solver does both.
3. Line search on a merit function or a *filter* for a step length α_k : $\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{d}_k^{\mathbf{x}}$, $\lambda_{k+1} = \lambda_k + \alpha_k \mathbf{d}_k^{\lambda}$.
4. Once the barrier problem is solved to ε_μ , decrease μ and repeat.

The Software Stack



Modeling Layers

	Interface	Derivatives
AMPL (Fourer, n.d.)	standalone language	AMPL Solver Library
Pyomo (Hart et al., 2017)	Python	via .nl/ASL, or its own
JuMP (Lubin et al., 2023)	Julia	expression-graph reverse mode
CasADi (Andersson et al., 2019)	Python/MATLAB/C++	symbolic + code generation
JAX	Python	dense AD; sparse ∇g , $\nabla^2 \mathcal{L}$
PyTorch	Python	via coloring + matrix-vector products
ExaModels (Shin, 2025)	Julia	GPU-focused: one kernel/pattern

Solvers and Linear Solvers

NLP solvers

Ipopt	interior point	filter line search
KNITRO	interior point	trust region, and an active set
MadNLP	interior point	GPU-capable linear algebra

Sparse symmetric indefinite solvers

MA27 / MA57 / MA97	HSL	$\mathbf{LBL}^\top$, 1×1 and 2×2 pivots
Pardiso, MUMPS	parallel	the same, at larger scale
cuDSS	GPU	no 2×2 pivots : Cholesky, $\mathbf{LDL}^\top$, LU

The CPU solvers pivot, and get the inertia as a by-product. The GPU solver does not. **Closing that gap is Part III.**

Summary of Part I

1. A nonlinear program is characterized by the **KKT conditions**; the difficulty is the complementarity condition.
2. **Interior-point methods** smooth it with a barrier and follow the central path. Per iteration: one derivative evaluation, one **sparse symmetric indefinite factorization**, one **inertia** check that conventionally needs numerical pivoting.

That factorization (and the numerical pivoting it relies on) is the layer **Part III** has to move onto the device.

Part II

Optimal Control as a Structured NLP

Direct transcription, orthogonal collocation, structured NLPs

Outline

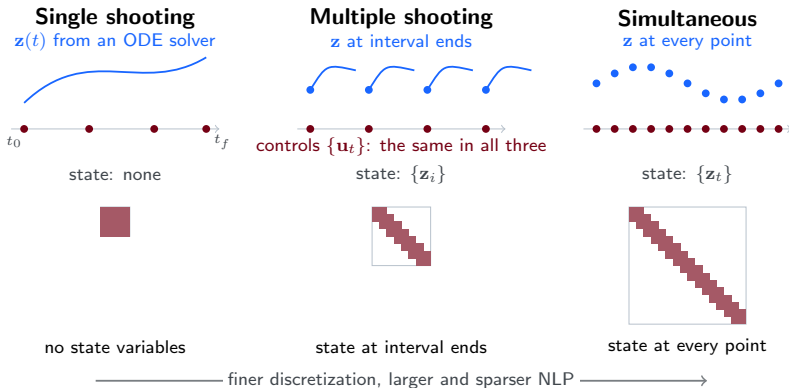
- ▶ Part I: Nonlinear Optimization
- ▶ **Part II: Optimal Control as a Structured NLP**
 - ▶ From Optimal Control to Nonlinear Programs
 - ▶ Collocation
 - ▶ Summary
- ▶ Part III: GPU Acceleration
- ▶ Part IV: Hands-On Sessions

The Continuous-Time Problem

$$\begin{aligned} \min_{\mathbf{z}(\cdot), \mathbf{u}(\cdot)} \quad & \varphi(\mathbf{z}(t_f)) + \int_{t_0}^{t_f} \psi(\mathbf{z}(t), \mathbf{u}(t)) dt \\ \text{s. t.} \quad & \dot{\mathbf{z}}(t) = \mathbf{f}(\mathbf{z}(t), \mathbf{u}(t)), \quad t \in [t_0, t_f] \\ & \mathbf{z}(t_0) = \mathbf{z}_0 \\ & \mathbf{g}(\mathbf{z}(t), \mathbf{u}(t)) \leq \mathbf{0}, \quad t \in [t_0, t_f] \end{aligned}$$

- ▶ The decision variables are *functions*: an infinite-dimensional problem.
- ▶ Two routes to a computable problem:
 - ▶ **Indirect**: “optimize then discretize”. Pontryagin gives a two-point boundary value problem.
 - ▶ **Direct**: “discretize then optimize”. Finitely many parameters give a finite-dimensional NLP.
- ▶ **This workshop is entirely about the direct route**: what all the software is built around.

Three Direct Approaches



- The simultaneous form suits a GPU because **the sequential ODE solve is gone**, replaced by algebraic constraints that evaluate in parallel, not because of the sparsity.
- What is given up is **adaptive time stepping**, which the shooting methods keep.

Running Example: The Goddard Rocket Problem

A rocket ascends vertically against drag and gravity; maximize the final altitude.

$$\begin{aligned} \max_{h,v,m,\tau} \quad & h(t_f) \\ \text{s. t.} \quad & \dot{h} = v \\ & \dot{v} = \frac{\tau - D(h,v)}{m} - g(h) \\ & \dot{m} = -\tau/c \\ & h(0) = h^0, \ v(0) = 0, \ m(0) = m^0 \\ & m(t_f) = m^f \end{aligned}$$

- States: altitude h , velocity v , mass m
- Control: thrust τ
- Drag $D(h,v) = D_c v^2 e^{-h_c(h-h^0)/h^0}$
- Gravity $g(h) = g_0(h^0/h)^2$

A classical benchmark from the COPS collection (Dolan and Moré, 2001); see also Betts (2010).

The Rocket, Transcribed

The simplest transcription: the trapezoidal rule on a uniform grid $t = 0, \dots, T$.

$$\begin{aligned} \max_{h_t, v_t, m_t, \tau_t, \Delta t} \quad & h_T \\ \text{s. t.} \quad & h_t = h_{t-1} + \frac{\Delta t}{2}(v_t + v_{t-1}) \\ & v_t = v_{t-1} + \frac{\Delta t}{2} \left(\frac{\tau_t - D(h_t, v_t)}{m_t} - g(h_t) + \frac{\tau_{t-1} - D(h_{t-1}, v_{t-1})}{m_{t-1}} - g(h_{t-1}) \right) \\ & m_t = m_{t-1} - \frac{\Delta t}{2c}(\tau_t + \tau_{t-1}) \\ & h_0 = h^0, \quad v_0 = 0, \quad m_0 = m^0, \quad m_T = m^f \end{aligned}$$

- ▶ A finite-dimensional NLP with $4(T+1) + 1$ variables and $3T + 4$ equality constraints, and nothing beyond that.
- ▶ Refining the discretization means increasing T : the problem grows, but **the algebra does not change**. Five distinct algebraic patterns, for any T .

Discretization Order and Truncation Error

	local error	accumulated
Euler	$\mathcal{O}(h^2)$	$\mathcal{O}(h)$
Trapezoidal	$\mathcal{O}(h^3)$	$\mathcal{O}(h^2)$
Collocation, K points	$\mathcal{O}(h^{2K+1})$	$\mathcal{O}(h^{2K})$

- ▶ What we want from a discretization: **accuracy**, **stability on stiff dynamics**, and a form that is *purely algebraic*.
- ▶ **Low order decays slowly**: halving h doubles N , so accuracy is bought with discretization points and the NLP is **large-scale by construction**.
- ▶ A fixed low-order rule is also **hard to keep stable** on stiff dynamics.
- ▶ So the standard practice is to buy accuracy from the **scheme** instead: **high order**, and specifically **orthogonal collocation**.

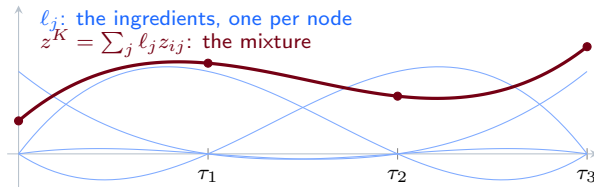
Chapter 10 of Biegler (2010).

Collocation: Polynomial Representation of the State

Represent the state on an element as a **linear combination of polynomials**:

$$z(t) \overset{\text{want}}{\approx} z^K(t) = \sum_{j=0}^K \ell_j(\tau) z_{ij}, \quad \text{the } z_{ij} \text{ are the unknowns.}$$

- ▶ We cannot make $\dot{z}^K = \mathbf{f}(z^K, u)$ hold at **every** t with finitely many unknowns.
- ▶ We therefore impose it at **K chosen points**, the **collocation points**, which defines the method.



Collocation on One Element

On element i of length h_i , with $\tau \in [0, 1]$, the state is a degree- K Lagrange polynomial through its values at $\tau_0 = 0 < \tau_1 < \dots < \tau_K$:

$$z^K(t) = \sum_{j=0}^K \ell_j(\tau) z_{ij}, \quad \ell_j(\tau) = \prod_{\substack{k=0 \\ k \neq j}}^K \frac{\tau - \tau_k}{\tau_j - \tau_k}.$$

Since $\ell_j(\tau_k) = \delta_{jk}$, the z_{ij} are the state values at the collocation points. Enforcing the ODE there gives the **collocation equations**

$$\sum_{j=0}^K D_{kj} z_{ij} = h_i \mathbf{f}(z_{ik}, u_{ik}), \quad D_{kj} = \frac{d\ell_j}{d\tau}(\tau_k).$$

precomputed constants

decision variables

- D is fixed, built once from the τ_k : each equation is a linear combination of unknowns equal to $h_i \mathbf{f}(\cdot)$, which is what an NLP solver takes.

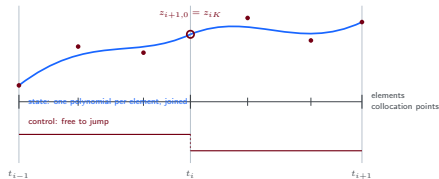
Connecting the Elements

Orthogonal collocation on finite elements

Split $[t_0, t_f]$ into N elements. On each, represent the state by a *polynomial* that satisfies the ODE *exactly* at K interior **collocation points**, and connect the polynomials across elements.

Each element is collocated **independently**, so nothing yet ties the pieces together:

$$z_{i+1,0} = \sum_{j=0}^K \ell_j(1) z_{ij}, \quad i = 1, \dots, N-1, \quad z_{1,0} = z_0.$$



The Resulting NLP

Collecting everything, the simultaneous formulation is

$$\begin{aligned} \min_{z_{ij}, u_{ik}, h_i} \quad & \varphi(z_f) + \sum_{i=1}^N h_i \sum_{k=1}^K \omega_k \psi(z_{ik}, u_{ik}) \\ \text{s. t.} \quad & \sum_{j=0}^K z_{ij} \dot{\ell}_j(\tau_k) = h_i \mathbf{f}(z_{ik}, u_{ik}), & i = 1, \dots, N, \quad k = 1, \dots, K \quad (\text{collocation}) \\ & z_{i+1,0} = \sum_{j=0}^K \ell_j(1) z_{ij}, \quad z_{1,0} = z_0 & i = 1, \dots, N-1 \quad (\text{continuity}) \\ & \mathbf{g}(z_{ik}, u_{ik}) \leq \mathbf{0} & i = 1, \dots, N, \quad k = 1, \dots, K \quad (\text{path}) \end{aligned}$$

precomputed constants

decision variables

- Size: $O((n_z K + n_u K)N)$ variables. Refining N or K makes it bigger, never harder in kind.
- Path constraints are enforced *pointwise at the collocation points*.

Biegler (2010, Eq. (10.19)).

Choosing the Collocation Points

- ▶ The τ_k are roots of a shifted Gauss–Jacobi polynomial, chosen to make the implied quadrature as accurate as possible.
- ▶ One construction, three **weight functions**. The weight decides whether the element endpoints are forced into the point set.
- ▶ **Legendre** takes neither: $O(h^{2K})$. **Radau** takes the right one, so $\tau_K = 1$: $O(h^{2K-1})$, with **stiff decay**. **Lobatto** takes both: $O(h^{2K-2})$.
- ▶ Radau is the usual default: stiff decay, and $\tau_K = 1$ makes the continuity condition trivial, since $\ell_j(1) = \delta_{jK}$ and consecutive elements simply **share the node**.

Truncation orders from Biegler (2010, Table 10.1).

Summary of Part II

1. **Direct transcription** turns an optimal control problem into an NLP, and **orthogonal collocation on finite elements** does it at high order, stably for stiff systems, and purely algebraically.
2. What comes out is **large, sparse, block-banded, and above all repetitive**: a fixed number of algebraic patterns evaluated over many data points.

Part III: the repetition buys us parallel derivative evaluation; the regular sparsity buys us GPU factorization, once we solve the pivoting problem.

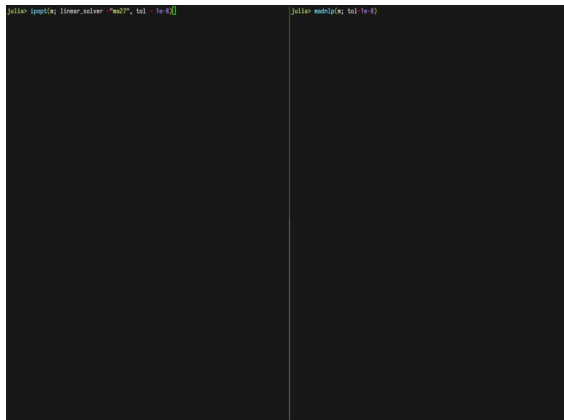
Part III

GPU Acceleration

SIMD abstraction, pivoting-free linear algebra, condensed-space IPM

MadNLP Converging on a GPU

AC optimal power flow, PGLIB-OPF case13659_pegase (13 659 buses), to 10^{-8} : Ipopt + MA27 on the CPU (left) against MadNLP on the GPU (right)



```
julia> Ipopt(; linear_solver "ma27", tol = 1e-8)
julia> madnlp(; tol = 1e-8)
```

Live solver logs during the solve.

Outline

- ▶ Part I: Nonlinear Optimization
- ▶ Part II: Optimal Control as a Structured NLP
- ▶ **Part III: GPU Acceleration**
 - ▶ A GPU Primer for Optimizers
 - ▶ Why NLP on GPUs Is Hard
 - ▶ SIMD Abstraction: ExaModels.jl
 - ▶ Pivoting-Free Linear Algebra
 - ▶ Condensed-Space Interior-Point Methods
 - ▶ Numerical Results
 - ▶ Variants and Outlook
- ▶ Part IV: Hands-On Sessions

Motivation: Optimization on Accelerated Hardware



CPU node 2 ×
AMD EPYC 9965
45 TFLOP/s FP64
1.2 TB/s memory

VS.



GPU node 8 ×
AMD Instinct MI325X
654 TFLOP/s FP64
48 TB/s memory

$\approx 14\times$ compute
 $\approx 39\times$ memory
per node

A GPU node offers **an order of magnitude** more compute and far more memory bandwidth, which is what puts **larger and harder optimization problems** in reach.

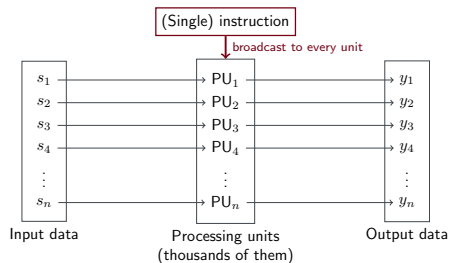
Currently the fastest CPU and GPU chips by peak **double-precision** (FP64) throughput, per [waredb.com/ranking/vector/fp64/all](https://www.waredb.com/ranking/vector/fp64/all). A node is taken as 2 sockets or 8 accelerators.

The Current Landscape

- ▶ **Factorization-free methods** have been the dominant approach on GPUs:
 - ▶ PDLP: first-order method for LPs (Applegate et al., 2022; Lu et al., 2025)
 - ▶ preconditioned CG within ADMM for convex QPs (Schubiger et al., 2020)
 - ▶ augmented Lagrangian + IPM + CG for nonlinear problems (Cao et al., 2016)
- ▶ They use only **GPU-friendly kernels** (spmv, axpy, map, mapreduce), which is why they worked first. They scale, but **high precision remains hard**, and dynamic optimization usually wants it.

Can we run second-order methods with direct linear algebra on GPUs?

The Execution Model: SIMD



- ▶ A CPU core executes a **complicated sequence of instructions serially**, and does so very efficiently. Its whole design is spent on finishing one such sequence quickly.
- ▶ A GPU keeps **thousands** of simple sequences running at once, all on the **same** instruction over different data: **SIMD**. Each is slow; there are very many.

Strengths and Weaknesses of the Hardware

Suitable

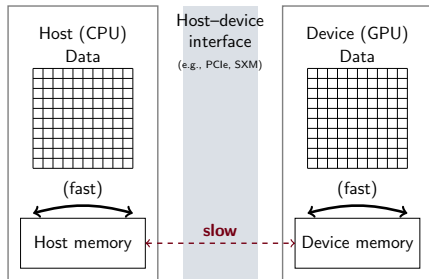
- ▶ **Uniform work:** one operation over many elements (map, a_{xy}, mapreduce, spmv).
- ▶ **Regular access:** neighboring threads, neighboring addresses.
- ▶ **Independent work:** dense linear algebra, batched small problems.

Unsuitable

- ▶ **Divergent work:** different paths in a group are serialized; decisions wait on the numbers.
- ▶ **Irregular access:** indirect, scattered addresses.
- ▶ **Sequential dependencies:** step $k + 1$ waiting on step k .



Host and Device Memory

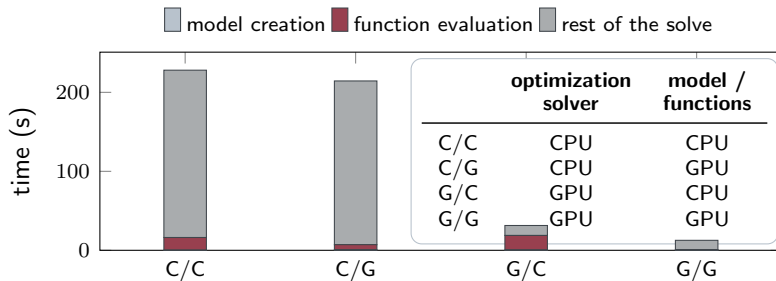


- ▶ Device memory has very high bandwidth to the device's own cores. The link to the host is **narrow and high-latency** by comparison.
- ▶ One round trip per iteration cancels the gain, which is why **partial porting fails**.

the data must live on the device, and so must the operations

The Cost of Partial Porting

AC optimal power flow, **PGLIB-OPF 78,478 buses**: model and solver on **CPU** or **GPU**



- ▶ Solver to the GPU: 228 s $\rightarrow$ 31 s. Model as well: 31 s $\rightarrow$ 13 s.
- ▶ With the model left on the host (G/C), **evaluation costs more than in the all-CPU solve**: every callback pays a transfer and a barrier.

Xeon 6767P + RTX PRO 6000 Blackwell; cf. Shin (2025).

Algorithmic Redesign for GPUs

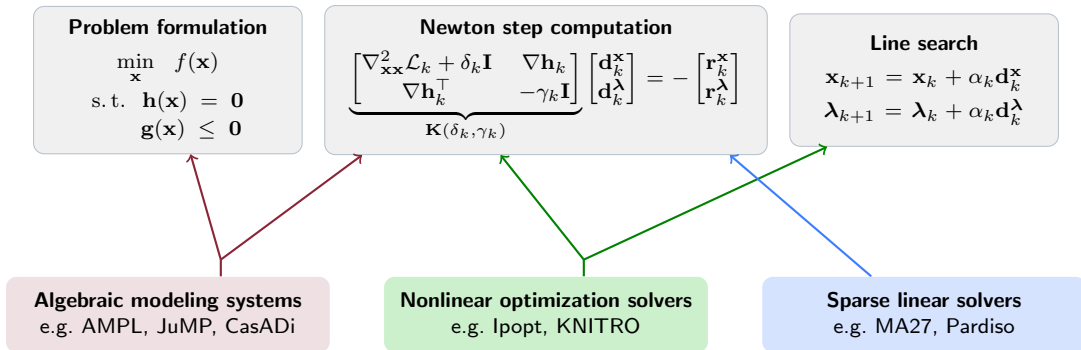
- ▶ A good CPU solver does not become a GPU solver by recompilation: the parts that make it good on a CPU (adaptive, sequential, data-dependent decisions) are the parts that do not run.
- ▶ So the useful question is not “how do I parallelize this loop?” but “**what equivalent problem can I solve instead, with no sequential decisions in it?**”

Adapting CPU code into GPU code is not merely a matter of software engineering.

The Three Components to Port to the GPU



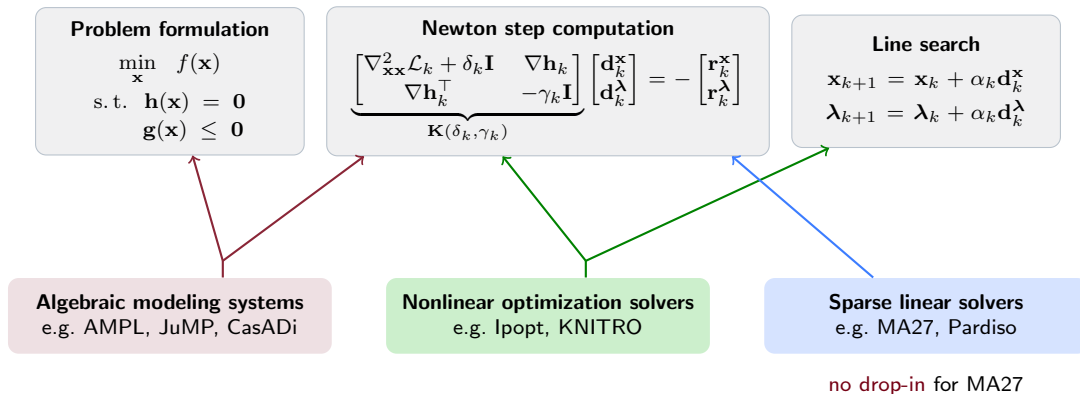
The same three layers from Part I. **Every one of them** has to move:



The Three Components to Port to the GPU



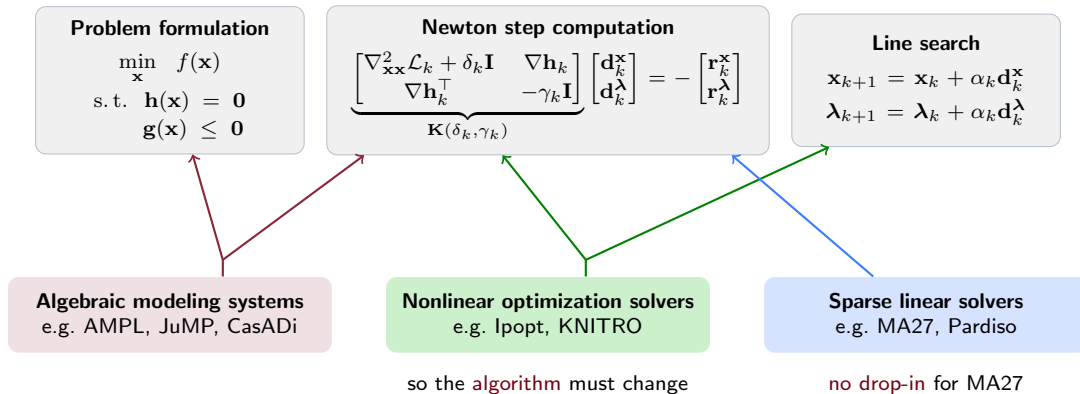
The same three layers from Part I. **Every one of them** has to move:



The Three Components to Port to the GPU



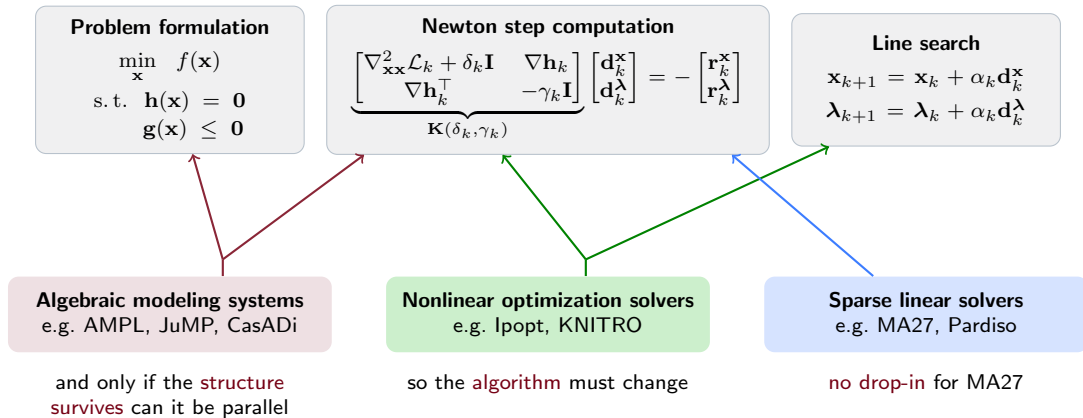
The same three layers from Part I. **Every one of them** has to move:



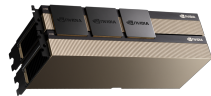
The Three Components to Port to the GPU



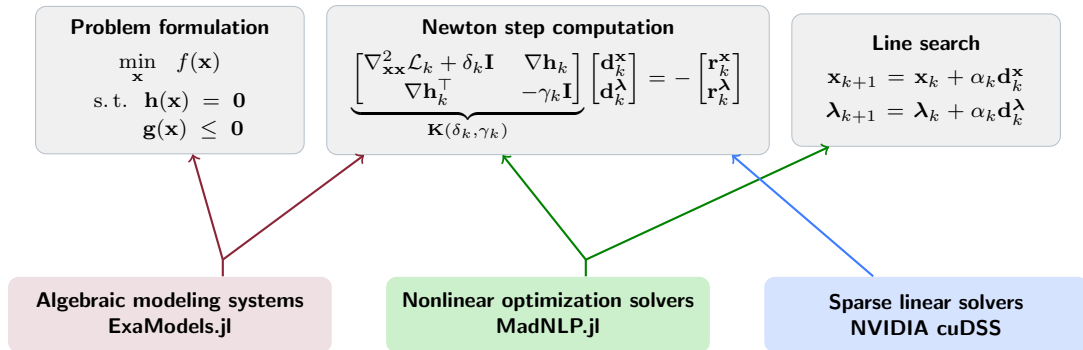
The same three layers from Part I. **Every one of them** has to move:



The Three Components to Port to the GPU



The same three layers from Part I. **Every one of them** has to move:



The SIMD Abstraction

Many large NLPs are a **small number of algebraic patterns**, instantiated many times.

$$\begin{aligned} & \max_{h_t, v_t, m_t, \tau_t, \Delta t} && h_T \\ \text{s. t.} &&& h_t = h_{t-1} + \frac{\Delta t}{2}(v_t + v_{t-1}), && t = 1, \dots, T \\ &&& v_t = v_{t-1} + \frac{\Delta t}{2} \left(\frac{\tau_t - D(h_t, v_t)}{m_t} - g(h_t) + \frac{\tau_{t-1} - D(h_{t-1}, v_{t-1})}{m_{t-1}} - g(h_{t-1}) \right), && t = 1, \dots, T \\ &&& m_t = m_{t-1} - \frac{\Delta t}{2c}(\tau_t + \tau_{t-1}), && t = 1, \dots, T \\ &&& h_0 = h^0, \quad v_0 = 0, \quad m_0 = m^0, \quad m_T = m^f \end{aligned}$$

The SIMD Abstraction

Many large NLPs are a **small number of algebraic patterns**, instantiated many times.

$$\begin{aligned} \min_{\mathbf{x}^b \leq \mathbf{x} \leq \mathbf{x}^\#} \quad & \sum_{l \in [L]} \sum_{i \in [I_l]} f^{(l)}(\mathbf{x}; p_i^{(l)}) \\ \text{s. t.} \quad & g^{(m)}(\mathbf{x}; q_j^{(m)}) = 0, \quad \forall m \in [M], j \in [J_m] \end{aligned}$$

The SIMD Abstraction

Many large NLPs are a **small number of algebraic patterns**, instantiated many times.

$$\begin{aligned} \min_{\mathbf{x}^b \leq \mathbf{x} \leq \mathbf{x}^\sharp} \quad & \sum_{l \in [L]} \sum_{i \in [I_l]} f^{(l)}(\mathbf{x}; p_i^{(l)}) \\ \text{s. t.} \quad & g^{(m)}(\mathbf{x}; q_j^{(m)}) = 0, \quad \forall m \in [M], j \in [J_m] \end{aligned}$$

- ▶ $f^{(l)}, g^{(m)}$: the **patterns**, a handful, known at compile time. $p_i^{(l)}, q_j^{(m)}$: the **data**, very many. A finer mesh adds data, **no new expression**.
- ▶ One tree walk and one compiled kernel per pattern, run over all the data **in parallel**.
Sparsity from the same tree: **no coloring**.

The Rocket in ExaModels

```
1 using ExaModels
2 backend = nothing # or CUDABackend() for the GPU
3 core = ExaCore(; backend, minimize = false)
4 @add_var(core, h, 0:T; lvar = h0, start = 1.0)
5 @add_var(core, v, 0:T; lvar = 0.0, start = 0.0)
6 @add_var(core, m, 0:T; lvar = mf, uvar = m0, start = m0)
7 @add_var(core, tau, 0:T; lvar = 0.0, uvar = tau_max, start = tau_max)
8 @add_var(core, dt, 1; lvar = 0.0, start = 1/T)
9
10 @add_obj(core, h[t] for t = T:T)
11 @add_con(core, h[t] - h[t-1] - 0.5*dt[1]*(v[t] + v[t-1]) for t = 1:T)
12 @add_con(core, m[t] - m[t-1] + 0.5*dt[1]/c*(tau[t] + tau[t-1]) for t = 1:T)
13 # ... velocity dynamics, initial and terminal conditions
14 model = ExaModel(core)
```

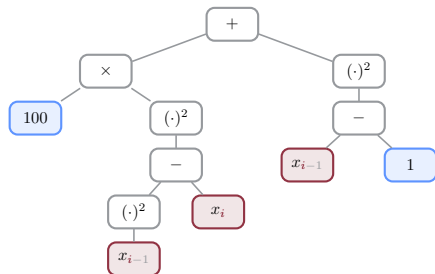
- ▶ The for stays **inside** the call. What comes before it is the **pattern**, what comes after is the **iterator**; they are stored separately and **never flattened**.
- ▶ A for loop *around* the call emits T unrelated objects, and the repetition is lost.

Shin, Schanen, et al. (2026); ExaModels.jl (Shin, 2025); abstraction from Shin, Anitescu, et al. (2024).

The Parametrized Expression Tree

One constraint or objective term of the model, as ExaModels stores it:

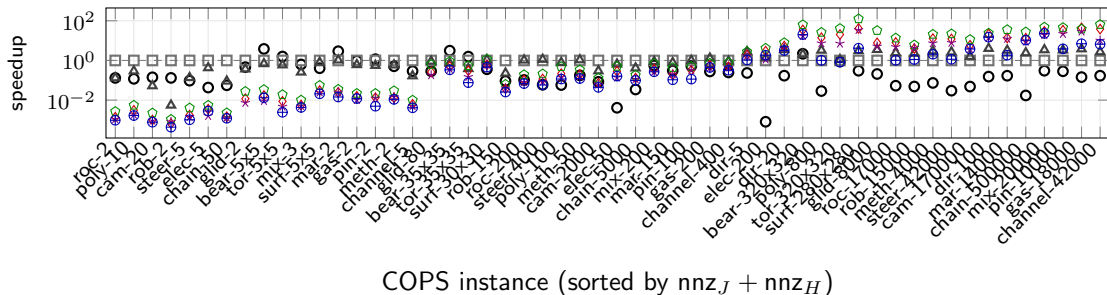
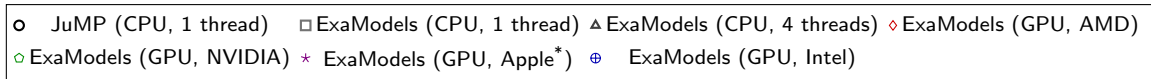
$$100 (x_{i-1}^2 - x_i)^2 + (x_{i-1} - 1)^2, \quad i = 2, \dots, N$$



- i is a **placeholder**: one tree for all $N - 1$ terms.
- **Derivative kernel and sparsity, once per tree**: evaluation is a parallel loop.

Shin, Schanen, et al. (2026); ExaModels.jl (Shin, 2025); abstraction from Shin, Anitescu, et al. (2024).

Performance of the Modeling Layer



Speedup over a single CPU thread on an **AMD EPYC 9474F**; COPS suite, per-instance evaluation times. * the Apple runs are single precision. (Shin, Schanen, et al., 2026)

The Fixed Cost of a Device Call

Hessian evaluation	Small (16) $\text{nnz} < 10^3$	Medium (18) $10^3 \leq \text{nnz} < 10^5$	Large (20) $10^5 \leq \text{nnz}$
CPU, 1 thread (EPYC 9474F)	$1.9 \mu\text{s}$	$50.2 \mu\text{s}$	2.83 ms
GPU (NVIDIA B200)	$71.9 \mu\text{s}$	$75.6 \mu\text{s}$	$92.9 \mu\text{s}$
GPU speedup	$0.0\times$	$0.7\times$	$30\times$

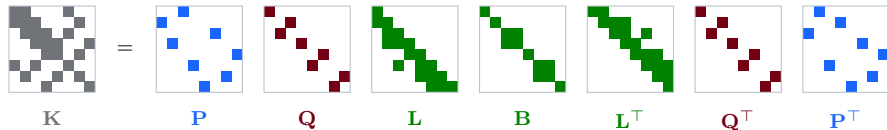
every interaction with the device, a transfer or a launch, has a fixed latency: make **few, large requests, not many small ones**

Data from Shin, Schanen, et al. (2026): ExaModels `hess_coord!`, COPS suite, shifted geometric mean per size class ($\sigma = 10^{-5}$ s).

Numerical Pivoting

From Part I: every iteration solves the KKT system, and **that solve is where the time goes**. It means **factorizing $\mathbf{K}$** , which in the sparse case carries two permutations:

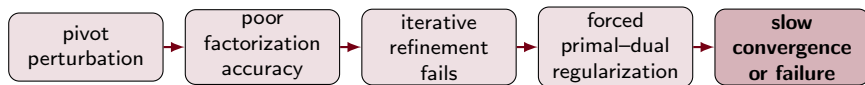
$$\mathbf{K} = \mathbf{P} \mathbf{Q} \mathbf{L} \mathbf{B} \mathbf{L}^T \mathbf{Q}^T \mathbf{P}^T$$



- **P**: fill-reducing reordering, from the sparsity pattern, computed once and reused.
- **Q**: numerical pivoting, chosen *on the fly*. **The obstacle**: irregular access, no parallelism.
- **B** is block diagonal, 1×1 and 2×2 : where the **inertia** is read off.

Pivot Perturbation and Pivoting-Free Classes

- ▶ GPU sparse direct solvers, cuDSS among them, **cannot do $\mathbf{LBL}^\top$** : no adequately robust numerical pivoting for indefinite systems.
- ▶ Run a vanilla IPM on one anyway and what you get is **pivot perturbation**, which is not free:



- ▶ Fine on small problems, **failing frequently beyond roughly 10^4 variables**.
- ▶ Two classes need **no pivoting at all**: **SPD** (Cholesky exists and is stable) and **SQD** (the factorization exists for *any* symmetric permutation).

Condensation

For a problem with **inequalities only**,

$$\min_{\mathbf{x}} f(\mathbf{x}) \quad \text{s. t.} \quad \mathbf{g}(\mathbf{x}) \leq \mathbf{0},$$

the KKT system of Part I, written with slacks, has **marked** blocks that are diagonal, so $\mathbf{d}_s$ and $\mathbf{d}_\lambda$ eliminate by exact substitution:

$$\begin{bmatrix} \nabla_{\mathbf{xx}}^2 \mathcal{L} + \delta_p \mathbf{I} & \nabla \mathbf{g}(\mathbf{x}) \\ \nabla \mathbf{g}(\mathbf{x})^\top & -\mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{d}_x \\ \mathbf{d}_s \\ -\mathbf{d}_\lambda \end{bmatrix} = - \begin{bmatrix} \nabla f(\mathbf{x}) - \nabla \mathbf{g}(\mathbf{x}) \boldsymbol{\lambda} \\ \boldsymbol{\Lambda} \mathbf{1} - \mu \mathbf{S}^{-1} \mathbf{1} \\ \mathbf{g}(\mathbf{x}) - \mathbf{s} \end{bmatrix}$$
$$\iff \underbrace{(\nabla_{\mathbf{xx}}^2 \mathcal{L} + \delta_p \mathbf{I} + \nabla \mathbf{g}(\mathbf{x}) (\delta_d \mathbf{I} + \boldsymbol{\Lambda}^{-1} \mathbf{S})^{-1} \nabla \mathbf{g}(\mathbf{x})^\top)}_{\mathbf{K}_{\text{cond}}} \mathbf{d}_x = -\mathbf{r}_{\text{cond}}$$

- ▶ Nothing is approximated, and $\mathbf{K}_{\text{cond}}$ is $n \times n$ rather than $(n + 2m) \times (n + 2m)$.
- ▶ It is **positive definite** exactly when the original system had the right inertia, so a successful Cholesky is the certificate, and **no pivoting is needed**.

Equality Constraints and Condensation

Optimal control problems are *made* of equality constraints: every collocation equation, every continuity condition.

$$\min_{\mathbf{x}} f(\mathbf{x})$$

$$\text{s. t. } \mathbf{h}(\mathbf{x}) = \mathbf{0}$$

$$\mathbf{g}(\mathbf{x}) \leq \mathbf{0}$$

$$\begin{bmatrix} \mathbf{K}_{\text{cond}} + \delta_p \mathbf{I} & \nabla \mathbf{h}(\mathbf{x}) \\ \nabla \mathbf{h}(\mathbf{x})^\top & -\delta_d \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{d}_x \\ \mathbf{d}_\lambda \end{bmatrix} = - \begin{bmatrix} \mathbf{r}_x \\ \mathbf{r}_\lambda \end{bmatrix}$$

- $\mathbf{h}(\mathbf{x}) = \mathbf{0}$ has **no slack**, so no diagonal block and no elimination: condensation leaves the equality block behind and the system stays **indefinite**.

Option 1: Lifted KKT

relax the equalities into narrow inequalities, then condense everything

Option 2: Hybrid KKT

keep the equalities, and handle the resulting Schur complement iteratively

Option 1: Lifted KKT

Step 1: lift and relax, with τ no larger than the solver tolerance:

$$\mathbf{h}(\mathbf{x}) = \mathbf{0} \quad \implies \quad \mathbf{h}(\mathbf{x}) + \mathbf{s} = \mathbf{0} \quad \text{and} \quad -\tau \leq \mathbf{s} \leq \tau$$

- ▶ Every constraint now has a slack, so condensation applies throughout and the condensed matrix is SPD under the usual inertia condition.
- ▶ cuDSS factorizes it by Cholesky on a pattern fixed in advance: no pivoting, no excessive regularization.

What you are paying

The answer solves a **relaxed** problem: as accurate as the tolerance you asked for, never more accurate than τ , and the condensed matrix grows ill-conditioned as $\mu \rightarrow 0$.

Option 2: Hybrid KKT

$$\begin{bmatrix} \mathbf{K}_{\text{cond}} + \delta_p \mathbf{I} & \nabla \mathbf{h}(\mathbf{x}) \\ \nabla \mathbf{h}(\mathbf{x})^\top & -\delta_d \mathbf{I} \end{bmatrix} \xrightarrow[\text{(solution unchanged)}]{+\rho \nabla \mathbf{h}(\mathbf{x}) \nabla \mathbf{h}(\mathbf{x})^\top} \begin{bmatrix} \mathbf{K}_{\text{cond}} + \delta_p \mathbf{I} + \rho \nabla \mathbf{h}(\mathbf{x}) \nabla \mathbf{h}(\mathbf{x})^\top & \nabla \mathbf{h}(\mathbf{x}) \\ \nabla \mathbf{h}(\mathbf{x})^\top & -\delta_d \mathbf{I} \end{bmatrix}$$

- ▶ The (1, 1) block, call it $\mathbf{K}_\rho$, is **SPD** for ρ large enough, iff the inertia is correct, so it is Cholesky-factorizable, once, without pivoting.
- ▶ Then solve the Schur complement in $\mathbf{d}_\lambda$ **iteratively** (CG):

$$\underbrace{\left(\nabla \mathbf{h}(\mathbf{x})^\top \mathbf{K}_\rho^{-1} \nabla \mathbf{h}(\mathbf{x}) + \delta_d \mathbf{I} \right)}_{\rightarrow \rho^{-1} \mathbf{I} \text{ as } \rho \rightarrow \infty} \mathbf{d}_\lambda = \mathbf{r}'_\lambda$$

- ▶ The limit is the mechanism: large ρ drives the Schur complement toward $\rho^{-1} \mathbf{I}$, so its spectrum clusters and CG converges in very few iterations, each a triangular solve plus a sparse product.
- ▶ **No relaxation** and **no accuracy ceiling**; the price is an inner iteration and a parameter ρ .

Golub and Greif (2003); Regev et al. (2023); Pacaud et al. (2024).

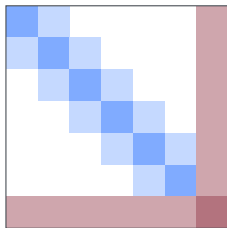
Lifted KKT and Hybrid KKT Compared

	Lifted KKT	Hybrid KKT
Equalities	relaxed to $\pm\tau$	kept exactly
System solved	one SPD system, $n \times n$	SPD system + CG on Schur complement
Linear solve	purely direct	direct + iterative
Accuracy	limited by τ	not limited
Extra parameter	τ	ρ
Main cost	ill-conditioning as $\mu \downarrow 0$	ill-conditioning as $\rho \uparrow$; CG iterations

- ▶ Both are implemented in **MadNLP.jl**, and both sit on cuDSS. The hands-on uses **Lifted KKT**, MadNLP's default on the GPU.
- ▶ Neither is uniformly better. Lifted KKT is the simpler and usually the faster at moderate tolerance; Hybrid KKT is what you reach for when you need tight convergence.

Block Tridiagonal Structure of the KKT Matrix

Transcribing a horizon couples stage t only to $t \pm 1$, so the KKT matrix is **block tridiagonal** (bordered, if there are variables shared across the horizon such as Δt):



blue: one block per stage, plus its couplings

red: the border, from horizon-wide variables

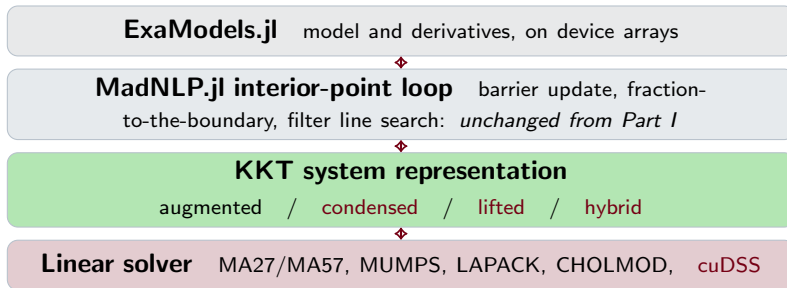
- ▶ A sequential sweep down the diagonal is the **Thomas algorithm**: N steps, no parallelism. **Cyclic reduction** eliminates every other stage and recurses: $\log N$ levels of **independent** block work.
- ▶ Each level is then many identical small dense factorizations, dispatched as a single batched call, which suits the hardware.

Parallel Block Tridiagonal Factorization

- ▶ cuDSS is **structure-agnostic**: given a matrix, not a horizon. A good ordering recovers much of the parallelism, but the schedule is **not exposed**.
- ▶ Exploiting the structure explicitly is a **live research direction**:
 - ▶ Jin et al. (2026): recursive Schur-complement reduction, **batched BLAS/LAPACK**, NVIDIA and AMD.
 - ▶ Schwan et al. (2026): nested dissection, Cholesky from $\mathcal{O}(Nn^3)$ to $\mathcal{O}(\log_2 N n^3)$.

**The first of those is in this congress: Wednesday 26 August,
14:30, Room 211, session Numerical Methods for Optimal Control.**

Implementation in MadNLP.jl



The COPS Benchmark

Tol	Solver	Small $\text{nnz} < 2^{18}$		Medium $2^{18} \leq \text{nnz} < 2^{20}$		Large $2^{20} \leq \text{nnz}$		Total	
		Solved	Time	Solved	Time	Solved	Time	Solved	Time
10^{-4}	MadNLP (GPU)	13	0.8665	15	4.8665	16	3.8194	44	3.2314
	Ipopt (CPU)	13	5.2315	15	15.9701	15	45.8411	43	19.2243
10^{-8}	MadNLP (GPU)	13	0.8575	16	1.5572	16	8.3549	45	3.3797
	Ipopt (CPU)	13	5.9413	15	17.6758	15	40.8639	43	19.2999

- **Benchmark:** COPS (Dolan and More, 2001), largely **dynamic optimization** problems, discretized as in Part II.
- **Hardware:** NVIDIA GV100 / Intel Xeon Gold 6130. **Time:** SGM10, 900 s limit.

Table reproduced from Montoison, Pacaud, Shin, et al. (2025).

Other Routes to a Pivoting-Free Solver

- ▶ Lifted KKT and Hybrid KKT are two points in a design space, not the only two.
- ▶ **MadNCL**: an augmented-Lagrangian algorithm. The quadratic penalty term is inherently GPU-friendly; it uses the nonlinearly-constrained Lagrangian formulation together with the condensation ideas above (Montoison, Pacaud, Saunders, et al., 2025).
- ▶ **MadIPM**: for convex quadratic programs; uses regularization to avoid pivoting altogether (Montoison, Pacaud, Shin, et al., 2025).
- ▶ Factorizing the augmented system directly, with no pivoting and no condensation, is under active investigation.
- ▶ Further out: multi-GPU for problems that exceed device memory, and mixed precision: factorize in single, refine in double.

Implications for Optimal Control

- ▶ **The simultaneous approach suits this hardware:** transcribing the whole horizon removes the sequential ODE solve and leaves something large, sparse, **block tridiagonal** and **repetitive**.
- ▶ **On COPS**, the largest instances go from **45.8 s to 3.8 s** at 10^{-4} , about **12×**, and the GPU solves 16 of them where the CPU solves 15.
- ▶ **The route is not settled:** Lifted KKT, Hybrid KKT, null-space and factorization-free methods all trade differently. Which wins for a given problem class remains open.

The software is installable today: ExaModels.jl, MadNLP.jl, cuDSS. What we would like from this workshop is for large-scale optimal control to start using it.

Part IV

Hands-On Sessions

Julia on the GPU; optimal control and parameter estimation end to end

Outline

- ▶ Part I: Nonlinear Optimization
- ▶ Part II: Optimal Control as a Structured NLP
- ▶ Part III: GPU Acceleration
- ▶ **Part IV: Hands-On Sessions**

References

- Andersson, Joel A. E. et al. (Mar. 2019). "CasADi: A Software Framework for Nonlinear Optimization and Optimal Control". In: *Mathematical Programming Computation* 11.1, pp. 1–36.
- Applegate, David et al. (Jan. 2022). *Practical Large-Scale Linear Programming Using Primal-Dual Hybrid Gradient*. arXiv: 2106.04756 [math].
- Betts, John T. (Jan. 2010). *Practical Methods for Optimal Control and Estimation Using Nonlinear Programming*. Second. Society for Industrial and Applied Mathematics.
- Biegler, Lorenz T. (2010). *Nonlinear Programming: Concepts, Algorithms, and Applications to Chemical Processes*. MOS-SIAM Series on Optimization 10. Philadelphia, Pa: SIAM [u.a.]
- Byrd, Richard H. et al. (2006). "Knitro: An Integrated Package for Nonlinear Optimization". In: *Large-Scale Nonlinear Optimization*. Ed. by G. Di Pillo and M. Roma. Boston, MA: Springer US, pp. 35–59.
- Cao, Yankai et al. (2016). "An Augmented Lagrangian Interior-Point Approach for Large-Scale NLP Problems on Graphics Processing Units". In: *Computers & Chemical Engineering* 85, pp. 76–83.
- Dolan, E. D. and J. J. Moré (Jan. 2001). *Benchmarking Optimization Software with COPS*. Tech. rep. ANL/MCS-TM-246. Argonne National Lab., IL (US).
- Duff, I. S. and J. K. Reid (Sept. 1983). "The Multifrontal Solution of Indefinite Sparse Symmetric Linear". In: *ACM Transactions on Mathematical Software* 9.3, pp. 302–325.
- Duff, Iain S. (June 2004). "MA57—a Code for the Solution of Sparse Symmetric Definite and Indefinite Systems". In: *ACM Trans. Math. Softw.* 30.2, pp. 118–144.
- Fourer, Robert (n.d.). "AMPL: A Mathematical Programming Language". In: ().
- Golub, Gene H. and Chen Greif (Jan. 2003). "On Solving Block-Structured Indefinite Linear Systems". In: *SIAM Journal on Scientific Computing* 24.6, pp. 2076–2092.
- Hart, William E. et al. (2017). *Pyomo — Optimization Modeling in Python*. Vol. 67. Springer Optimization and Its Applications. Cham: Springer International Publishing.
- Jin, David et al. (Aug. 2026). "Harnessing Batched GPU Kernels for Solutions of Block Tridiagonal Systems". In: *23rd IFAC World Congress*. Paper WeB13.5, session "Numerical Methods for Optimal Control"; preprint arXiv:2509.03015. Busan, Republic of Korea.
- Lu, Haihao et al. (July 2025). *cuPDLP+: A Further Enhanced GPU-Based First-Order Solver for Linear Programming*. arXiv: 2507.14051 [math].
- Lubin, Miles et al. (Sept. 2023). "JuMP 1.0: Recent Improvements to a Modeling Language for Mathematical Optimization". In: *Mathematical Programming Computation* 15.3, pp. 581–589.
- Montoison, Alexis, François Pacaud, Michael Saunders, et al. (Oct. 2025). *MadNCL: A GPU Implementation of Algorithm NCL for Large-Scale, Degenerate Nonlinear Programs*. arXiv: 2510.05885 [math].
- Montoison, Alexis, François Pacaud, Sungho Shin, et al. (Nov. 2025). "GPU Implementation of Second-Order Linear and Nonlinear Programming Solvers". In: *NeurIPS Workshop on GPU-Accelerated and Scalable Optimization*.

References II

- Nocedal, Jorge and Stephen J. Wright (2006). *Numerical Optimization*. 2nd ed. Springer Series in Operations Research. New York: Springer.
- NVIDIA (2024). *NVIDIA cuDSS (Preview): A High-Performance CUDA Library for Direct Sparse Solvers — NVIDIA cuDSS Documentation*.
<https://docs.nvidia.com/cuda/cudss/index.html>.
- Pacaud, François et al. (May 2024). *Condensed Interior-Point Methods for Scalable Nonlinear Programming on GPUs*. To appear in *Mathematical Programming Computation*. arXiv: 2405.14236 [math].
- Regev, Shaked et al. (Mar. 2023). "HyKKT: A Hybrid Direct-Iterative Method for Solving KKT Linear Systems". In: *Optimization Methods and Software* 38.2, pp. 332–355.
- Schubiger, Michel et al. (Oct. 2020). "GPU Acceleration of ADMM for Large-Scale Quadratic Programming". In: *Journal of Parallel and Distributed Computing* 144, pp. 55–67.
- Schwan, Roland et al. (Jan. 2026). *GPU-Accelerated Cholesky Factorization of Block Tridiagonal Matrices*. arXiv: 2601.03754 [math.OC].
- Shin, Sungho (Feb. 2025). *ExaModels.JI*. Exanauts.
- Shin, Sungho, Mihai Anitescu, et al. (Nov. 2024). "Accelerating Optimal Power Flow with GPUs: SIMD Abstraction of Nonlinear Programs and Condensed-Space Interior-Point Methods". In: *Electric Power Systems Research* 236, p. 110651.
- Shin, Sungho, François Pacaud, et al. (Apr. 2025). *MadNLP.JI*. MadNLP.
- Shin, Sungho, Michel Schanen, et al. (Aug. 2026). *ExaModels.JI: An Algebraic Modeling System for Nonlinear Programming on GPUs*. arXiv: 2608.16265 [math.OC].
- Świrydowicz, Kasia et al. (July 2022). "Linear Solvers for Power Grid Optimization Problems: A Review of GPU-accelerated Linear Solvers". In: *Parallel Computing* 111, p. 102870.
- Vanderbei, Robert J. (Feb. 1995). "Symmetric Quasidefinite Matrices". In: *SIAM Journal on Optimization* 5.1, pp. 100–113.
- Wächter, Andreas and Lorenz T. Biegler (Mar. 2006). "On the Implementation of an Interior-Point Filter Line-Search Algorithm for Large-Scale Nonlinear Programming". In: *Mathematical Programming* 106.1, pp. 25–57.